

Frobenius Separation of Massive and Massless Sectors in $GF(27)^*$

Algebraic Structure of the $\tilde{T} \cdot m = 1$ Transition
and Connection to 8 Gluons

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Frobenius Separation of Massive and Massless Sectors in $GF(27)^*$

Abstract

The multiplicative group $GF(27)^* \cong \mathbb{Z}_{26}$ decomposes under the Frobenius automorphism $\phi : x \mapsto x^3$ into exactly two types: two fixed points $\{+1, -1\}$ and eight three-element orbits. Via the decomposition $\mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ (CRT), four $\mathbb{Z}_{13}$ three-orbits times two $\mathbb{Z}_2$ parities yield the eight gluons of the adjoint $SU(3)$ representation. The orbit decomposition itself is algebraically forced. For the fixed points $\{+1, -1\}$ there are two readings: as the massive $\mathbb{Z}_2$ sector (particle/antiparticle) and, as $GF(3)^*$ of the fixed field $GF(3)$, as the $U(1)$ sector (photon). Since it is the same set, this double assignment is not resolved; the assignment is open **[S]**. Whether the massless modes carry spin 1 does not follow from the Frobenius structure (Doc. 231, open). Status: **[K]**.

.1 Starting Point

In Doc. 338

$$\left(\frac{m_\mu}{m_e}\right)^2 = 43200 = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| = 64 \cdot 25 \cdot 27$$

is shown for the bare mass ratio as an algebraically exact relation **[B]**. The fundamental relation $\tilde{T} \cdot m = 1$ describes a duality: massive sector ($m > 0$, compact torus S_m^1) and massless sector ($m \rightarrow 0$, $\tilde{T} \rightarrow \infty$). In Doc. 285 the $3'$ extension is identified as a kinetic (massless) window. This note asks: what algebraic structure separates both sectors within $GF(3) \subset GF(9)$, $GF(27) \subset GF(729)$? ($GF(9)$ is not a subfield of $GF(27)$, since $2 \nmid 3$; the only subfields of $GF(27)$ are $GF(3)$ and $GF(27)$, and the smallest common extension of $GF(9)$ and $GF(27)$ is $GF(729)$.)

.2 Decomposition of $\mathbb{Z}_{26}$ under the Frobenius

Basic structure **[B]**

$GF(27)^* \cong \mathbb{Z}_{26}$, cyclic of order $3^3 - 1 = 26 = 2 \cdot 13$. Since $\gcd(2, 13) = 1$:

$$\mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}.$$

The Frobenius $\phi : x \mapsto x^3$ acts:

- On $\mathbb{Z}_2$: $3 \equiv 1 \pmod{2}$ — ϕ is *trivial*.
- On $\mathbb{Z}_{13}$: $\text{ord}(3, \mathbb{Z}_{13}) = 3$ (since $3^3 = 27 \equiv 1 \pmod{13}$) — ϕ has order 3.

Orbits of ϕ on $\mathbb{Z}_{13}$ **[B]**

Type	Elements
Fixed point	$\{0\}$
Three-orbit 1	$\{1, 3, 9\}$
Three-orbit 2	$\{2, 5, 6\}$
Three-orbit 3	$\{4, 10, 12\}$
Three-orbit 4	$\{7, 8, 11\}$

Complete orbit table in $\mathbb{Z}_{26}$ **[B]**

$\mathbb{Z}_2$ parity	$\mathbb{Z}_{13}$ orbit	$\mathbb{Z}_{26}$ orbit
0	$\{0\}$	$\{0\}$ (fixed point)
1	$\{0\}$	$\{13\}$ (fixed point)
0	$\{1, 3, 9\}$	$\{14, 16, 22\}$
1	$\{1, 3, 9\}$	$\{1, 3, 9\}$
0	$\{2, 5, 6\}$	$\{2, 6, 18\}$
1	$\{2, 5, 6\}$	$\{5, 15, 19\}$
0	$\{4, 10, 12\}$	$\{4, 10, 12\}$
1	$\{4, 10, 12\}$	$\{17, 23, 25\}$
0	$\{7, 8, 11\}$	$\{8, 20, 24\}$
1	$\{7, 8, 11\}$	$\{7, 11, 21\}$

Check: $2 + 8 \times 3 = 26 = |GF(27)^*|$. ✓

.3 Physical Identification [K]

Massive sector: fixed points $= \mathbb{Z}_2$

The two fixed points $\{+1, -1\}$ form the massive sector: every massive particle has an antiparticle ($\mathbb{Z}_2$ structure), consistent with the compact torus sector.

Massless $SU(3)$ sector: eight gluons

$$8 \text{ gluons} = 4 \mathbb{Z}_{13}\text{-orbits} \times 2 \mathbb{Z}_2\text{-parities} = |GF(9)^*|.$$

The three elements of each $\mathbb{Z}_{13}$ orbit correspond to the three colours $\{R, G, B\}$; the $\mathbb{Z}_2$ parity distinguishes colour from anticolour. This is the adjoint representation of $SU(3)$, derived from Galois structure alone.

$U(1)$ sector: photon from the fixed field $GF(3)$

The fixed field of the Frobenius on $GF(27)$ is $GF(3)$ (Fermat: $x^3 = x$ for $x \in GF(3)$). The projection $GF(27) \rightarrow GF(3)$ gives:

Structure	Sector
$GF(3)^* = \{1, 2\} \cong \mathbb{Z}_2$	Photon: $U(1)$, own antiparticle
$27 - 3 = 24$ remaining elements	$SU(3)$ sector (quarks, gluons)

$GF(3)^* = \{1, 2\}$ is the same set as the fixed points $\{+1, -1\}$ of the previous section. The assignment of this set to the photon and to the massive sector is not resolved and remains open [S].

.4 Algebraic Necessity [B]

$$\text{Non-fixed elements in } GF(27)^* = 26 - 2 = 24, \quad (1)$$

$$24 = 8 \cdot 3 = |GF(9)^*| \cdot |\text{colours}| = (3^2 - 1) \cdot 3 = 3^3 - 3, \quad (2)$$

$$8 \text{ gluons} = \frac{26-2}{3} = 8. \quad (3)$$

All three relations follow solely from $|GF(27)| = 3^3$ and the order 3 of the Frobenius on $\mathbb{Z}_{13}$.

.5 Complete Picture

Galois structure	Physical sector	Symmetry
$GF(3)^* = \mathbb{Z}_2$	Photon	$U(1)$
Fixed points $\{+1, -1\} \subset GF(27)^*$	Massive particle/antiparticle	$\mathbb{Z}_2$
4×2 three-orbits in $GF(27)^*$	8 gluons	$SU(3)_{\text{adj}}$

The first two rows concern the same set $GF(3)^* = \{+1, -1\}$; which of the two assignments holds is open [S].

The relation $8 \times 3 = 24 = \text{Kissing}(D_4)$ is algebraically forced ($3^3 - 3 = 24$). Whether $\text{Kissing}(D_4)$ has a deeper geometric connection to this Galois structure is subject to further investigation.

Verification script: `pruef_339_frobenius.py` (6/6 assertions).

.6 Scales of the Duality and Effective Masslessness [K]

The two poles of $\tilde{T} \cdot m = 1$

The fundamental relation $\tilde{T} \cdot m = 1$ has two excluded edges (Doc. 306), both finite and quantified:

Pole	Length	Time	Energy per cell
Dense (derived)	$L_0 = \xi \ell_P$	$T_0 = \xi t_P$	$E_{\max} = E_P/\xi \approx 9.2 \times 10^{22} \text{ GeV}$
Thin (calibrated)	$R_H = \hbar/E_H$	$T_H = \hbar/E_H$	$E_H = E_0 \xi^{41/4} \approx 1.4 \times 10^{-33} \text{ eV}$

At both ends the reciprocity closes exactly: $T_0 \cdot E_{\max} = t_P \cdot E_P = 1$ and $T_H \cdot E_H = 1$ (natural units). There is no infinity — neither above nor below. The dense pole is derived from ξ [B]; the thin pole is calibrated to the measured H_0 (Doc. 182, P35/P36) [S].

What “massless” means in FFGFT

$\tilde{T} \cdot m = 1$ has no solution at $m = 0$ — FFGFT knows no empty state, only a thinnest one (Doc. 306). The minimal rest mass per cell is:

$$m_0 = \frac{E_H}{c^2} \approx 1.4 \times 10^{-33} \text{ eV}/c^2.$$

Gluons and photons are modes in the limit $m \rightarrow m_0$, not at $m = 0$. The time circle S_m^1 opens to the time axis $\mathbb{R}_t$ in this limit (type-III pullback, Doc. 332, R96).

The deviation of the dispersion relation from the massless formula $\omega = |\mathbf{k}|$ is:

$$\frac{\delta\omega}{\omega} = \frac{m_0^2}{2k^2} \approx \begin{cases} 2.5 \times 10^{-67} & k = 2 \text{ eV (visible light)} \\ 1.7 \times 10^{-59} & k = 2.4 \times 10^{-4} \text{ eV (CMB)} \\ 9.8 \times 10^{-85} & k = 1 \text{ GeV (QCD)} \end{cases}$$

The experimental upper bound on the photon rest mass is $m_\gamma < 10^{-18} \text{ eV}/c^2$ (PDG). The FFGFT minimum mass m_0 lies 10^{15} times below this — beyond any conceivable measurement.

Conclusion

“Massless” in FFGFT is not an algebraic exactness ($m \neq 0$) but a structural statement: the rest mass lies below the Hubble energy scale E_H/c^2 and is indistinguishable from zero for all physical purposes. Both poles of the duality are finite and quantified — there is no infinity, but both limits lie far beyond any experimental reach.

.7 Frobenius Structure, Masslessness and Spin [K]

Derivation

The derivation proceeds in four steps.

Step 1 [K]: The Frobenius map $\phi : x \mapsto x^3$ on $GF(27)^*$ corresponds in the type-III pullback picture (Doc. 332, R96) to the conformal scaling of the wave vector:

$$k \mapsto 3k, \quad \omega(3k) = 3\omega(k).$$

This follows directly from the massless dispersion relation $\omega = c|k|$. Massive modes do not satisfy this relation: for $m > 0$, $\omega^2 = c^2 k^2 + (mc^2/\hbar)^2$, so $\omega(3k) \neq 3\omega(k)$. Fixed points transform trivially: $\phi(x) = x$ corresponds to $k \mapsto k$.

Step 2 [B, standard result]: Wigner (1939) classifies the representations of the Poincaré group by m^2 and little group; massless particles are distinguished by their helicity h . The linear dispersion $\omega(\lambda k) = \lambda \omega(k)$ characterises masslessness for every spin $(0, \frac{1}{2}, 1, \dots)$ and does not fix the spin. Massive fields do not satisfy it (in the non-relativistic limit $k \rightarrow 0$ one has $\omega \rightarrow mc^2/\hbar$).

Step 3 [K]: The 8 three-orbits of $GF(27)^*$ carry the conformal scaling $k \mapsto 3k$ and therefore correspond to massless modes; that they are spin-1 fields would have to be shown via the winding numbers on $L^2(T^3, S)$ (Doc. 231, open). The 2 fixed points $\{+1, -1\}$ transform trivially and correspond to the massive sector.

Step 4 [K]: The fixed field $GF(3)^* = \{1, 2\} \cong \mathbb{Z}_2$ is conformally invariant (fixed under the projection $GF(27) \rightarrow GF(3)$) and carries a massless singlet, identified with the photon; spin 1 is open (Doc. 231). Step 3 places the fixed points $\{+1, -1\}$ in the massive sector; Step 4 and the table in the section on the $U(1)$ sector assign the same set $GF(3)^* = \{1, 2\} = \{+1, -1\}$ to the photon. This double assignment is not resolved; the assignment is open [S].

Compatibility with the FFGFT corpus

In FFGFT, spin arises from winding numbers (n_θ, n_ϕ) on the spin bundle $L^2(T^3, S)$ (Doc. 231), not from an external Poincaré representation. The Wigner classification in Step 2 is therefore a standard result [B] that is consistent with the FFGFT spin-bundle framework but does not replace it. A massless spin-1 field has the two helicities ± 1 (three states only in the massive case); an assignment to the three-orbits of $GF(27)^*$ under $\mathbb{Z}_3$ therefore does not follow directly and remains as an open bridge to Doc. 231.

Result

Frobenius type	Sector	Spin	Physical object
Fixed points $\{+1, -1\}$	Massive, $k \mapsto k$	–	Leptons, quarks
8 three-orbits	Massless, $k \mapsto 3k$	open	8 gluons
Fixed field $GF(3)^*$	Massless, $k \mapsto k$	open	Photon (singlet)

Note: The spin of the massive fixed points (leptons: $\frac{1}{2}$, quarks: $\frac{1}{2}$) does not follow from the Frobenius structure but from the winding numbers on $L^2(T^3, S)$ (Doc. 231). Likewise, spin 1 of the gluons and the photon has to be shown via Doc. 231. The first and third rows concern the same set $\{+1, -1\} = GF(3)^$; their assignment is open.*

.8 Context: Connection to HLV and the Hilbert Space Doubling [K]

The fork from Doc. 307

Doc. 307 poses the fundamental question: is time a parameter *beside* the state space (unrolled picture) or a compact dimension *inside* it (rolled-up picture)? Both descriptions use the same operator formalism; the difference lies in where time is located.

The rolled-up picture ($\tilde{T} \cdot m = 1$, $L^2(T^4) \otimes \mathbb{C}^3$) is the massive side: discrete modes, winding numbers (n_θ, n_ϕ) , lepton masses from $\xi = 4/30000$. The unrolled picture ($L^2(\mathbb{R}_t)$) is the massless

side — and exactly there, as the Frobenius decomposition of $GF(27)^*$ shows algebraically, the gluons live.

FFGFT, HLV and the doubling $3 \rightarrow 6$

In Doc. 285 (FFGFT-HLV dimensional bridge) the doubling $\mathbf{6} = \mathbf{3} \oplus \mathbf{3}'$ is specified from the Hilbert space perspective:

- **FFGFT sector ($\mathbf{3}$, compact torus S_m^1):** $\mathbb{Z}_3$ -crystallographic, 3-fold symmetry, lives natively in 3D/4D. Protected invariant: $\theta = 2/9$ (Koide), $r = \sqrt{2}$ ($Q = 2/3$). This is the massive sector.
- **HLV sector ($\mathbf{3}'$, type-III pullback to $\mathbb{R}_t$):** A_5 -icosahedral, 5-fold symmetry, requires 6D embedding. The unrolling referred to here is the lossless mathematical type (isometry $L^2(S_m^1) \cong \text{Per}_{R_m}(\mathbb{R}_t)$, Doc. 332 R96, type III), *not* the measurement act (type I, lossy, Doc. 333). Protected invariant: φ (golden ratio), $r = 2/\sqrt{5}$. This is the massless candidate.

The 6D splitting is forced by the crystallographic restriction: 5-fold symmetry is forbidden in 3D ($2 \cos 72^\circ = 1/\varphi \notin \mathbb{Z}$) but exists in 6D. The $\mathbf{3}'$ sector is geometrically present in FFGFT but physically empty without dynamics.

What the Frobenius decomposition contributes to the bridge [K]

The Frobenius decomposition of $GF(27)^*$ provides a new algebraic precision that Doc. 285 did not yet have:

Structure	Hilbert space picture	Galois picture
Massive sector ($\mathbf{3}$)	$L^2(S_m^1)$, rolled-up	Fixed points $\{+1, -1\}$ of ϕ
Massless sector ($\mathbf{3}'$)	$L^2(\mathbb{R}_t)$, type-III pullback (R96)	8 three-orbits of ϕ
Koide $\theta = 2/9$	$\mathbb{Z}_3$ circulant	$GF(9)^*$ structure (Doc. 338)
Golden ratio φ	A_5 icosahedron	not anchored in $GF(27)^*$

The golden ratio φ does not appear in the $\mathbb{Z}_{13}$ picture: $\mathbb{Z}_{13}$ carries no 5-fold symmetry (since $5 \nmid 12$). This is consistent with φ being the HLV invariant, not the FFGFT invariant.

The Frobenius structure of $GF(27)^*$ confirms algebraically what Doc. 285 shows geometrically: FFGFT is the $\mathbb{Z}_3$ -crystallographic core (massive fixed points), HLV would be the A_5 sector (massless $\mathbf{3}'$ orbits).

.9 Extension: Two Pictures, One Structure [K]

Unrolled picture and conformal symmetry

The Frobenius map $\phi : x \mapsto x^3$ on $GF(27)^*$ corresponds in the unrolled picture ($\mathbb{R}_t$ instead of S_m^1) to a conformal scaling of the wave vector:

$$k \mapsto 3k, \quad \omega \mapsto 3\omega.$$

This is consistent with the massless dispersion relation $\omega = c|k|$:

$$\omega(3k) = c \cdot |3k| = 3 \cdot c \cdot |k| = 3\omega(k). \quad \checkmark$$

For massive modes, $\omega^2 = c^2 k^2 + (mc^2/\hbar)^2$ gives $\omega(3k) \neq 3\omega(k)$ — massive modes are *not* conformally invariant. The Frobenius structure thus explains the separation algebraically:

Frobenius type	Physical sector	Symmetry
Fixed points $\{+1, -1\}$	Massive, $\omega(3k) \neq 3\omega(k)$	no conformal inv.
Three-orbits (8 total)	Massless, $\omega(3k) = 3\omega(k)$	conformal inv.

Accessible calculations per picture

Content	Picture	Status
Lepton masses, winding numbers (n_θ, n_ϕ)	Compact	[B]
Galois identity $43200 = 64 \cdot 25 \cdot 27$	Compact	[B]
$K_{\text{frak}}^{-36} \approx 16/\pi^2$	Compact	[B]
Gluon dispersion $\omega = c k $	Unrolled	[K]
Frobenius = conformal scaling $k \mapsto 3k$	Unrolled	[K]
Hilbert space isometry $L^2(S_m^1) \cong \text{Per}_{R_m}(\mathbb{R}_t)$	Both	[B]
$\xi = 4/30000$, mass ratios	Both	[B]

The conformal dimension of gluons ($\Delta = 1$, spin-1 vector bosons) remains to be derived from the FFGFT field theory — noted as an open point.

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.10 Note on Authorship

This document was developed on 29 August 2026 in collaboration with the AI assistant Claude (Anthropic, Sonnet 4.6). The algebraic computations (Frobenius orbits, $\mathbb{Z}_2 \times \mathbb{Z}_{13}$ decomposition, conformal symmetry test) were carried out by the assistant and verified by the author. The physical interpretation and all status markers (**[B]**, **[K]**) are the author's responsibility.

.11 Notation

[ASSUMPTION]	Axiom or declared external input – not derived
[K]	Core – derived from ξ and the assumptions, numerically verified
[B]	Bridge – algebraically proved
[S]	Sketch – plausible, not yet fully worked out